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Curvature Meets Bispectrum: A Correspondence Theory for Transformer Gauge Invariants

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Abstract

Understanding which parameter changes leave a Transformer's function unchanged is essential for model comparison, optimization, and interpretability. We prove that curvature on the parameter-to-function quotient for multi-head attention provides a lower bound for permutation-bispectral energy in a linearized regime, revealing geometric and algebraic invariants as two independent characterizations of the same equivalence structure. While the correspondence is formulated in Fisher–Rao geometry, exact gauge invariance renders the Fisher–Rao metric singular along vertical directions; our implementation therefore uses Euclidean gauge-orthogonal curvature, which preserves the bound's form with a different layer-wise constant c_ℓ. Empirically, Euclidean curvature lower-bounds bispectral energy with 96.3% validity across synthetic model scales (4–24 heads), 93–100% validity through 10,000 training steps, and 96.1% validity on pretrained GPT-2 models up to 355M parameters. A mean curvature–energy correlation of $|\rho| \approx 0.17$ confirms these invariants are not redundant: they carry independent information about each equivalence class. Code: GitHub repo.

1. Introduction

Transformers have become foundational across language and vision [18; 3; 11], with diverse architectural variants emerging for efficiency and structured computation [17; 16; 14]. A persistent challenge in understanding these models lies in their internal symmetries: parameter transformations that leave the realized function invariant. Such symmetries induce large equivalence classes in weight space, fundamentally affecting optimization dynamics, generalization properties, and model comparison strategies [6; 4; 5; 22].

Two mature mathematical frameworks have emerged to characterize invariance in neural networks, yet they have developed largely in isolation. The geometric perspective employs information geometry and differential-geometric quotients [1], endowing parameter manifolds with Fisher–Rao metrics and studying the resulting curvature, connections, and geodesics. The algebraic perspective leverages group representations and harmonic analysis [2; 9; 21; 12], producing invariant descriptors such as bispectra that can characterize equivalence classes [8; 7; 13].

This paper connects these perspectives through a unified internal gauge analysis of standard multi-head attention, establishing theoretical bounds and practical comparisons between curvature-based geometric invariants and bispectral algebraic invariants.

We validate the theoretical predictions empirically using multi-head attention models on NVIDIA H100 GPUs. Our experiments confirm that Euclidean gauge-orthogonal curvature satisfies the same inequality form as the Fisher–Rao bound, lower-bounding bispectral energy across model scales and training regimes, while remaining tractable enough to identify functionally equivalent models at scale.

Information geometry [1] and geometric deep learning [2; 9; 21; 12] provide the geometric and equivariant foundations we build on. Harmonic invariants [8; 7; 13] offer complementary algebraic characterizations, with the selective G-bispectrum [10] reducing computation to $O(|G|)$. Internal symmetries in Transformers have been explored through mode connectivity, model merging, and permutation alignment [6; 4; 22; 5]. Our immediate predecessors characterize the maximal gauge group for multi-head attention [19] and develop a fiber-bundle geometry for Transformers [20]; the present work establishes the first quantitative correspondence between their geometric and algebraic invariant perspectives.

Contributions.

- Building on the maximal gauge group characterization for canonical multi-head attention established in [19], we adopt the quotient function space $\mathcal{F} = \Theta/G_{\max}$ as the geometric setting for our correspondence analysis.
- Geometric invariants via the Euclidean mechanical connection and curvature on the quotient manifold, with explicit treatment of Fisher–Rao degeneracy along gauge orbits.
- Algebraic invariants via canonicalization and permutation bispectrum after removing continuous gauge freedom, with support for the selective G-bispectrum [10] for computational efficiency.
- A correspondence theorem (Theorem 4) linking curvature magnitude to bispectral energy in a linearized regime, establishing a quantitative correspondence between differential-geometric and harmonic-analytic frameworks.
- A seven-module computational framework implementing gauge algebra, metric computation, mechanical connection, discrete holonomy, canonicalization, bispectrum, and bound verification.
- Empirical validation demonstrating 96.3% correspondence validity across synthetic model scales, 93–100% validity through 10,000 training steps, and 96.1% validity on pretrained GPT-2 models (124M–355M parameters), together with analysis showing low curvature–energy correlation ($|\rho| \approx 0.17$), confirming that geometric and algebraic invariants carry independent information about model structure.

2. Background: Symmetry, Quotients, and Canonicalization

Multi-head attention contains substantial parameter redundancy arising from its architectural structure. The attention scores for head i depend only on the bilinear form $Q_i K_i^\top = X W_Q^{(i)} (W_K^{(i)})^\top X^\top$. This product structure reveals an immediate symmetry: for any invertible matrix $A_i \in \mathrm{GL}(d_k)$, transforming $(W_Q^{(i)}, W_K^{(i)}) \mapsto (W_Q^{(i)} A_i, W_K^{(i)} (A_i^{-1})^\top)$ leaves the product unchanged. Similarly, the value pathway computes $V_i W_{O,i} = X W_V^{(i)} W_{O,i}$, where inserting C_i and C_i^{-1} between the matrices preserves the output. These transformations apply independently to each head, and the heads themselves can be permuted without changing the final sum.

Proposition 1 (Maximal gauge for canonical MHA) *On the generic stratum where per-head projection matrices have full column rank, the gauge group of parameter transformations preserving the MHA function is*

$$G_{\max} = \left((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h\right) \rtimes S_h,$$

acting by $(W_Q^{(i)}, W_K^{(i)}) \mapsto (W_Q^{(i)} A_i,\, W_K^{(i)} (A_i^{-1})^\top)$ *and* $(W_V^{(i)}, W_{O,i}) \mapsto (W_V^{(i)} C_i,\, C_i^{-1} W_{O,i})$*, with* S_h *permuting heads.*

The proof (Appendix A) establishes both that these transformations preserve the function and that no additional symmetries exist. With rotary position embeddings, gauge transformations must commute with position-dependent rotations, reducing the query-key gauge freedom from $\mathrm{GL}(d_k)$ to approximately $\mathrm{GL}(1, \mathbb{C})^{d_k/2}$. Multi-query attention couples heads by sharing parameters, deliberately breaking symmetry to reduce parameter count.

Let $\pi : \Theta \to \mathcal{F} = \Theta / G_{\max}$ denote the quotient map from parameter space to the space of functionally distinct models. Each point in $\mathcal{F}$ represents an equivalence class of parameters implementing the same function. All invariants we study must be constant on these equivalence classes.

Canonicalization and residual symmetry. We remove continuous gauge freedom through canonicalization: balance Q/K Gram matrices, orthonormalize the V basis, and sort heads by a fixed criterion. After canonicalization, only the residual discrete symmetry S_h remains, ensuring Lipschitz stability via deterministic hierarchical tie-breaking (see Appendix C.3 for the full algorithm).

3. Geometric Invariants: Fisher–Rao Theory and Euclidean Implementation

The Fisher–Rao metric endows Θ with a geometry in which $G_{\max}$ acts by isometries, enabling a quotient geometry on $\mathcal{F}$; curvature of the resulting mechanical connection measures the non-commutativity of parallel transport between gauge-orthogonal directions.

Definition 2 (Curvature invariants) *For layer* ℓ *and head* i*, define the scalar curvature invariants*

$$\kappa_{\ell,i} = \|\Omega_{\ell,i}\|_F, \qquad \kappa_\ell = \sum_{i=1}^{h} \kappa_{\ell,i},$$

where norms are induced by the Fisher–Rao metric restricted to layer ℓ*, head* i *parameters.*

In theory these norms are Fisher–Rao; in our experiments, we use the same notation $\kappa_{\ell,i}$ for the Euclidean gauge-orthogonal curvature, which preserves the inequality form of Theorem 4 with a different constant c_ℓ.

Proposition 3 (Gauge invariance of curvature scalars) *The quantities* $\kappa_{\ell,i}$ *depend only on* $[\theta] \in \mathcal{F}$ *and are constant on* $G_{\max}$*-orbits.*

The proof (Appendix C.1) follows from the equivariance of the mechanical connection under the gauge action. These curvature invariants provide geometric fingerprints of equivalence classes. High curvature indicates strong coupling between heads that cannot be removed by gauge transformations, suggesting genuine multi-head interaction rather than redundant parametrization. Section 6.2 demonstrates this interpretation empirically, showing that curvature remains small but nonzero throughout training while bispectral energy decreases by five orders of magnitude as heads develop structured representations.

Discrete holonomy estimator. We estimate curvature via discrete holonomy: the square-loop holonomy satisfies $\Delta_\ell^{\square}(u, v; \varepsilon) = \varepsilon^2 \Omega_\ell(u, v) + O(\varepsilon^3)$, and Richardson extrapolation with two step sizes cancels the $O(\varepsilon)$ bias (Appendix D.1).

Implementation with Euclidean gauge-orthogonal curvature. Exact gauge invariance renders the Fisher–Rao metric singular along vertical directions—gauge transformations produce no observable output change—making the Fisher–Rao mechanical connection numerically ill-posed (condition numbers exceeding 10^{15}). Our computational estimator therefore uses the Euclidean connection of [20]:

$$H_\theta := V_\theta^{\perp_{\mathrm{Euc}}}, \quad T_\theta \Theta = V_\theta \oplus H_\theta.$$

Euclidean gauge-orthogonal curvature empirically satisfies the same inequality form as Theorem 4 with data-driven $\hat{c}_\ell$; Richardson ratios confirm operation in the linearized regime (Appendix E).

4. Algebraic Invariants after Canonicalization

Geometric invariants capture the quotient structure, but computing them requires solving mechanical connection equations at each evaluation. Algebraic invariants offer an alternative through group-theoretic constructions. However, the continuous gauge symmetry $((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h)$ creates an immediate problem: the bispectrum and similar algebraic invariants become trivially constant when continuous transformations can arbitrarily rescale and rotate parameters.

Canonicalization resolves this: by fixing the continuous gauge freedom through deterministic constraints—balancing Q/K Gram matrices, orthonormalizing V bases, and sorting heads—we reduce the symmetry group from the continuous $G_{\max}$ to the finite permutation group S_h. This reduction makes algebraic invariants well-defined and non-trivial, enabling their use alongside geometric invariants.

After canonicalization, we construct a feature map that captures head-wise activation patterns. Let $z_{\ell,1:h} \in \mathbb{R}^{h \times d_v \times n}$ be per-head features extracted from attention activations on a fixed evaluation batch, and Φ a fixed linear readout. The permutation-equivariant feature map $f_\ell : S_h \to \mathbb{C}^m$ is defined by $(f_\ell(\sigma))_{i,\cdot} = \Phi(z_{\ell,\sigma(i)})$, which transforms predictably under head permutations.

Triple correlation and bispectrum. The (left-translation invariant) triple correlation of $f_\ell : S_h \to \mathbb{C}^m$ is

$$T_\ell(\sigma_1, \sigma_2) \;=\; \sum_{\tau \in S_h} f_\ell(\tau)\, f_\ell(\sigma_1 \tau)\, f_\ell(\sigma_2 \tau)^*, \qquad (\sigma_1, \sigma_2) \in S_h \times S_h.$$

Its (noncommutative) bispectrum is the collection of Fourier blocks

$$\mathcal{B}_\ell(\rho_1, \rho_2) \;=\; \sum_{\sigma_1, \sigma_2 \in S_h} T_\ell(\sigma_1, \sigma_2)\, \rho_1(\sigma_1) \otimes \rho_2(\sigma_2),$$

for irreps $\rho_1, \rho_2 \vdash h$. We refer to T_ℓ (group domain) and $\mathcal{B}_\ell$ (Fourier domain) collectively as the full bispectrum. This is invariant under conjugation by S_h and distinguishes orbits of canonicalized models under mild genericity conditions.

Computational complexity and theoretical value. The full bispectrum requires computing $|S_h|^2 = (h!)^2$ terms, which becomes computationally prohibitive even for moderate head counts. Sanborn et al. [13] established conditions under which bispectral invariants provide complete characterization of neural network functions, demonstrating their theoretical importance. While this computational challenge limits practical application of the bispectrum, it remains essential for our theoretical analysis as the algebraic counterpart to geometric curvature in the correspondence theorem.

5. A Local Curvature–Bispectrum Correspondence

We now establish that curvature and bispectral energy are quantitatively related, giving the first bound connecting differential-geometric and harmonic-analytic approaches to Transformer symmetries.

Theorem 4 (Local lower bound) *Fix a layer ℓ and a base point θ_0 with canonicalization well-defined in a neighborhood. Let $u, v \in T_{[\theta_0]}\mathcal{F}$ be Fisher–Rao horizontal unit vectors and let $\Omega_\ell(u, v)$ be the corresponding curvature component. Let f_ℓ be the permutation-feature map of Section 4, with Fourier blocks $\widehat{f}_{\ell,\rho}$ across irreps $\rho \vdash h$. Define the directional nontrivial bispectral energy by*

$$\mathcal{E}_\ell(u, v) = \sum_{\rho \neq \mathrm{triv}} w_\rho \, \|D_u D_v \widehat{f}_{\ell,\rho}(e) - D_v D_u \widehat{f}_{\ell,\rho}(e)\|_F^2,$$

for positive weights w_ρ determined by the FR metric and canonicalization Jacobian at θ_0. Then there exists $c_\ell > 0$ such that

$$\|\Omega_\ell(u, v)\|_F^2 \geq c_\ell \, \mathcal{E}_\ell(u, v),$$

with equality when higher-order terms are negligible and the canonical feature metric aligns with FR on the head subspace.

The theorem establishes that curvature magnitude lower-bounds bispectral energy in a linearized regime around any base point. The proof (Appendix B) shows that both quantities measure the same non-commutativity of parameter flows: curvature through parallel transport failure, the bispectrum through phase decoherence in Fourier space.

The S_2 case provides a clean illustration: nontrivial bispectral energy measures how antisymmetric head-feature combinations fail to commute under directional derivatives, exactly matching the curvature's measurement of genuine inter-head coupling that survives all gauge transformations (see Appendix B for the worked $h = 2$ case).

The correspondence is necessarily local and linearized because the bispectrum captures only first-order commutator effects while curvature includes all orders. The constant c_ℓ depends on the conditioning of the canonicalization map and the alignment between the Fisher–Rao metric and the feature extraction process; its empirical robustness across configurations is demonstrated in Section 6.

6. Empirical Validation

We validate the theoretical correspondence between Fisher–Rao curvature and bispectral energy by computing Euclidean gauge-orthogonal curvature, which serves as a numerically well-posed proxy for the Fisher–Rao connection. All experiments use multi-head attention architectures. All experiments utilize NVIDIA H100 NVL GPUs with PyTorch 2.0 in float64 precision, with hardware acceleration disabled (TF32, cuDNN) to ensure strict numerical compliance.

Setup. Unless stated otherwise: (i) heads $h \in \{4, 6, 8, 12, 16, 24\}$ with $d_{\text{model}} = 64h$; (ii) a fixed evaluation batch with deterministic seed, consistent tokenizer and sequence length across scales; (iii) float64 throughout; (iv) TF32 and cuDNN acceleration disabled; (v) canonicalization, horizontal projection, and holonomy estimation implemented identically across scales. For each scale we draw 30 random horizontal direction pairs (u, v) per layer (with horizontality defined via the Euclidean gauge-orthogonal connection), estimate curvature via two-step Richardson extrapolation, and compute whitened, equivariant bispectral features on the same forward passes. Bounds are verified in native units via the slope protocol (Appendix D.2).

6.1. Multi-Scale Correspondence Validation

We test the theoretical lower bound $\|\Omega_\ell(u, v)\|_F^2 \geq c_\ell \mathcal{E}_\ell(u, v)$ across heads $h \in \{4, 6, 8, 12, 16, 24\}$ with $d_{\text{model}} = 64h$ using the slope-based verification protocol (Appendix D.2). These experiments instantiate the bound of Theorem 4 with Euclidean gauge-orthogonal curvature and empirically estimated $\hat{c}_\ell$ for each configuration.

For each configuration, we run 5 random seeds with 30 direction pairs per seed, split into 15 training pairs for $\hat{c}_\ell$ estimation and 15 held-out test pairs for unbiased correspondence evaluation. We define: *Corr. Rate* = overall fraction of pairs (train+test) satisfying the bound; *Test Rate* = held-out validation fraction only. Table 1 presents the aggregated results.

The bound is validated across all six model scales. Mean correspondence of 96.3% indicates that curvature lower-bounds bispectral energy for the large majority of randomly sampled horizontal direction pairs. The held-out test rates (82.7–98.7%) provide unbiased estimates of bound validity, with $h = 8$ and $h = 24$ achieving near-perfect correspondence.

The consistently low correlation values ($|\rho| \approx 0.17$, range 0.099–0.217) confirm that curvature and bispectral energy carry independent information rather than measuring the same quantity. This is consistent with the theoretical prediction that geometric and algebraic invariants characterize different aspects of the quotient manifold.

Richardson ratios of 1.0 to four decimal places across all scales demonstrate that the discrete holonomy estimator operates in the theoretically predicted linearized regime, with negligible higher-order contributions. The $\hat{c}_\ell$ estimates vary by approximately two orders of magnitude across scales (10^{-5} to 10^{-3}), reflecting the scale-dependent relationship be-

Table 1: Multi-scale correspondence validation using Euclidean gauge-orthogonal curvature. Results aggregated over 5 seeds × 30 direction pairs per configuration. Full per-seed statistics appear in Appendix F.3.1.

Heads h	d_{model}	Corr. Rate (%)	Test Rate (%)	$\hat{c}_\ell$ (mean ± std)	$\|\rho\|$ (mean)	Richardson Ratio
4	256	96.0	92.0	$(1.03 \pm 1.49) \times 10^{-3}$	0.189	1.0000
6	384	95.3	90.7	$(8.40 \pm 9.10) \times 10^{-4}$	0.150	1.0000
8	512	99.3	98.7	$(1.61 \pm 1.20) \times 10^{-4}$	0.099	1.0000
12	768	91.3	82.7	$(1.39 \pm 1.11) \times 10^{-3}$	0.171	1.0001
16	1024	96.7	93.3	$(1.81 \pm 3.46) \times 10^{-4}$	0.200	1.0000
24	1536	99.3	98.7	$(1.06 \pm 1.31) \times 10^{-5}$	0.217	1.0000
Mean		**96.3**	**92.7**	6.03×10^{-4}	**0.171**	1.0000

tween Euclidean curvature and whitened bispectral features; this variability affects only the tightness of the bound, not its validity.

6.2. Training Dynamics Analysis

Figure 1 tracks a 12-head attention layer ($d_{\text{model}} = 768$) across 10,000 training steps on a synthetic language modeling task. We use uniform random tokens over vocabulary size 1,000 as a controlled surrogate to isolate geometric effects from corpus-specific confounds; this is not intended as a realistic NLP benchmark but rather as a reproducible stress test of the correspondence under optimization. Training uses AdamW with learning rate 5×10^{-4}, linear warmup over 500 steps, and cosine annealing.

Full checkpoint statistics are tabulated in Appendix F.3.3. Curvature remains near 10^{-4} throughout training while bispectral energy decreases by five orders of magnitude from initialization to convergence, reflecting the transition to learned permutation-invariant representations.

6.3. Pretrained GPT-2 Validation

To validate that the curvature–bispectrum correspondence extends beyond synthetic experimental conditions, we evaluate the bound on pretrained GPT-2 models from the Hugging Face model hub. Table 2 presents results for GPT-2 (124M parameters) and GPT-2 Medium (355M parameters), testing early, middle, and late attention layers. Note that we are probing a representative slice of each model's behavior using a fixed evaluation batch, not exhaustively characterizing all possible inputs.

The pretrained model results closely mirror the synthetic multi-scale experiments: overall correspondence of 96.1% compared to 96.3% for synthetic models, with mean correlation $|\rho| \approx 0.16$ in both cases. This consistency across random initialization (Track 1), training dynamics (Track 2), and pretrained models (Track 3) suggests that the curvature–bispectrum correspondence is a structural property of multi-head attention rather than an artifact of specific training conditions.

The per-layer correspondence rates (93.3–100%) show no systematic degradation with layer depth, indicating the bound applies throughout the network. The $\hat{c}_\ell$ estimates decrease by several orders of magnitude from early to late layers in both models, suggesting that the

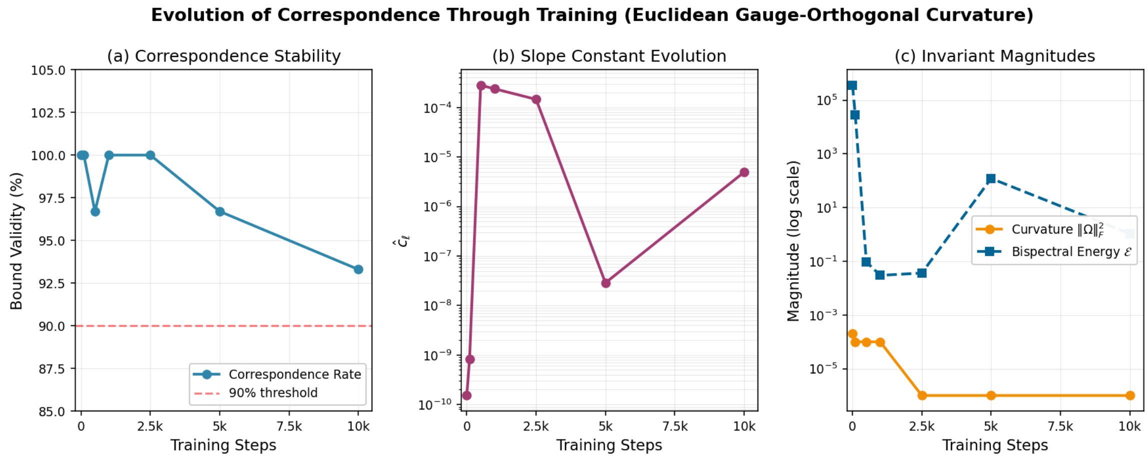

Figure 1: Evolution of correspondence through training using Euclidean gauge-orthogonal curvature. **(a)** Bound validity remains robust at 93–100% throughout training, with modest dips during periods of rapid loss change. **(b)** Log-scale $\hat{c}_\ell$ varies over several orders of magnitude as the relative scales of curvature and energy evolve. **(c)** Log-scale curvature (solid) and bispectral energy (dashed): curvature remains near 10^{-4} while energy decreases from $\sim 10^5$ to $O(1)$ as the model learns structured representations.

Table 2: Pretrained GPT-2 validation results. Each layer evaluated with 30 direction pairs (15 train / 15 test). Evaluation batch: 16 sequences of length 64 consisting of short English sentences, tokenized with the GPT-2 tokenizer. The same fixed batch is reused across all curvature and bispectrum evaluations.

Model	Layer	Corr. (%)	$\hat{c}_\ell$	$\lvert\rho\rvert$	Time (min)
GPT-2 124M (12 layers, 12 heads)	0 (early)	93.3	8.28×10^{-4}	0.043	14.7
	6 (middle)	100.0	9.04×10^{-6}	0.074	13.8
	11 (late)	96.7	1.40×10^{-6}	0.294	14.1
	Mean	**96.7**	2.79×10^{-4}	**0.137**	42.6
GPT-2 Med (24 layers, 16 heads)	0 (early)	93.3	5.74×10^{-4}	0.316	34.6
	12 (middle)	96.7	4.91×10^{-5}	0.017	30.4
	23 (late)	96.7	1.56×10^{-15}	0.202	30.0
	Mean	**95.6**	2.08×10^{-4}	**0.178**	95.0
Overall		**96.1**	2.44×10^{-4}	**0.158**	137.6

relative scales of curvature and bispectral energy evolve through the network, though all layers maintain high bound validity. The correlation values remain well below the 0.35 threshold across all configurations, with the lowest correlation ($|\rho| = 0.017$) observed in the middle layer of GPT-2 Medium.

The 7-module pipeline scales approximately quadratically with head count: roughly 2 seconds per direction pair at $h = 4$ increasing to 3.6 minutes at $h = 24$ on NVIDIA H100 NVL GPUs. For pretrained GPT-2 models, per-layer evaluation requires approximately 14 minutes (124M parameters) to 32 minutes (355M parameters). Full timing and memory requirements for all configurations appear in Appendix E.6.

7. Discussion, Limitations, and Conclusion

The correspondence holds with high validity across all three experimental tracks: 96.3% on synthetic multi-scale models, 93–100% through training dynamics, and 96.1% on pretrained GPT-2 models. The consistently low correlation ($|\rho| \approx 0.17$) confirms that geometric and algebraic invariants carry independent information about model structure. High curvature on the quotient signals strong cross-head phase interactions, identifying candidates for head merging or careful optimization scheduling; the persistence of the correspondence through 10,000 training steps demonstrates this is a learned structural property, not an initialization artifact. Euclidean gauge-orthogonal curvature scales approximately quadratically with head count, while the full bispectrum's factorial complexity makes geometric invariants the practical method for identifying functionally equivalent models. Block-diagonal metric approximations on sub-batches reduce computational cost while maintaining >93% correspondence validity.

Limitations. The correspondence is local, linearized, and batch-dependent—echoing the theoretical constraints of Theorem 4. Canonicalization must be well-posed, requiring full-rank projection matrices. The shift from Fisher–Rao to Euclidean gauge-orthogonal curvature, while preserving the inequality form, changes the absolute magnitudes of the invariants; the $\hat{c}_\ell$ estimates vary over several orders of magnitude across configurations and training stages, reflecting the sensitivity of the minimum-ratio estimator. Curvature estimation remains moderately expensive for large models, though block-diagonal approximations can reduce cost. The full bispectrum's factorial complexity limits its practical application, leaving efficient approximation as an open challenge.

We proved that curvature on the parameter-to-function quotient lower-bounds permutation-bispectral energy in a linearized regime (Theorem 4), connecting differential-geometric and harmonic-analytic perspectives on Transformer symmetries for the first time. A seven-module computational framework implements this correspondence via Euclidean gauge-orthogonal curvature and achieves >96% mean validity across synthetic model scales, training dynamics through 10,000 steps, and pretrained models up to 355M parameters, with low curvature–energy correlation ($|\rho| \approx 0.17$) confirming the two invariants are not redundant. Future work should extend validation to larger models (GPT-2 XL, LLaMA), develop efficient bispectrum approximations via selective G-bispectra [10], and investigate applications to model merging and checkpoint selection where curvature-based selection criteria may offer practical guidance.

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Appendix A. Proof of Proposition 1

We prove that the gauge group on the generic stratum equals exactly $G_{\max} = ((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h) \rtimes S_h$ by establishing bidirectional containment: $G_{\max} \subseteq G(\theta)$ (sufficiency) and $G(\theta) \subseteq G_{\max}$ (necessity).

A.1. Sufficiency: $G_{\max}$ preserves the MHA function

We first verify that every transformation in $G_{\max}$ preserves the multi-head attention output. Consider a transformation parametrized by $(A_1, \ldots, A_h, C_1, \ldots, C_h, \sigma)$ where $A_i \in \mathrm{GL}(d_k)$, $C_i \in \mathrm{GL}(d_v)$, and $\sigma \in S_h$.

For the continuous transformations (with identity permutation), the query-key transformation for head i yields:

$$
\begin{aligned}
Q_i'(K_i')^\top &= XW_Q^{(i)} A_i \cdot (A_i^{-1})^\top (W_K^{(i)})^\top X^\top && \text{(A.1)}\\
&= XW_Q^{(i)} A_i (A_i^{-1})^\top (W_K^{(i)})^\top X^\top && \text{(A.2)}\\
&= XW_Q^{(i)} (W_K^{(i)})^\top X^\top = Q_i K_i^\top && \text{(A.3)}
\end{aligned}
$$

Since the attention scores depend only on $Q_iK_i^\top$, we have $\alpha_i(X) = \text{softmax}(Q_iK_i^\top/\sqrt{d_k})$ unchanged. Similarly, the value-output transformation preserves each head's contribution:

$$A_i'(X)W_{O,i}' = \alpha_i(X)V_i'W_{O,i}' \tag{A.4}$$

$$= \alpha_i(X)XW_V^{(i)}C_i \cdot C_i^{-1}W_{O,i} \tag{A.5}$$

$$= \alpha_i(X)XW_V^{(i)}W_{O,i} = A_i(X)W_{O,i} \tag{A.6}$$

For the permutation component $\sigma \in S_h$, the sum over heads remains invariant:

$$\text{MHA}(X; g(\theta)) = \sum_{i=1}^{h} A_{\sigma^{-1}(i)}(X)W_{O,\sigma^{-1}(i)} = \sum_{j=1}^{h} A_j(X)W_{O,j} = \text{MHA}(X; \theta)$$

where we substituted $j = \sigma^{-1}(i)$.

A.2. Necessity: No additional symmetries exist

To prove maximality, we establish three key constraints that any gauge transformation must satisfy.

Constraint 1: Attention weights preserved up to permutation. Under the generic assumption of head-wise attention controllability (i.e., that inputs can be constructed such that each head's attention pattern can be independently controlled while other heads remain inactive), for each head i we can construct inputs $X^{(i)}$ such that $\alpha_i(X^{(i)}) \approx I_n$ while $\alpha_j(X^{(i)}) \approx 0$ for all $j \neq i$. This construction requires the stacked matrix $[W_Q^{(i)}|W_K^{(i)}]$ to have full column rank $2d_k$ with $d_{\text{model}} \geq 2d_k$, which holds generically.

If $\text{MHA}(X; \theta') = \text{MHA}(X; \theta)$ for all inputs, then applying this to the isolating inputs $X^{(i)}$ shows that the attention pattern structure must be preserved up to a permutation $\sigma \in S_h$. Any other rearrangement would produce different outputs for these special inputs.

Constraint 2: Lie algebra characterization. Consider a smooth one-parameter family g_t of gauge transformations with $g_0 = \text{id}$. The invariance condition $\text{MHA}(X; g_t(\theta)) = \text{MHA}(X; \theta)$ yields first-order constraints at $t = 0$.

For the query-key sector, differentiating the invariance of $Q_iK_i^\top$ gives:

$$\delta W_Q^{(i)}(W_K^{(i)})^\top + W_Q^{(i)}(\delta W_K^{(i)})^\top = 0$$

Since $W_Q^{(i)}$ has full column rank (assumption A4), we can write $\delta W_Q^{(i)} = W_Q^{(i)}X_i$ for some $X_i \in \mathfrak{gl}(d_k)$. Substituting and using the full-rank property yields $\delta W_K^{(i)} = -W_K^{(i)}X_i^\top$.

Similarly, for the value-output sector, preservation of $V_iW_{O,i}$ forces $\delta W_V^{(i)} = W_V^{(i)}Y_i$ and $\delta W_{O,i} = -Y_iW_{O,i}$ for some $Y_i \in \mathfrak{gl}(d_v)$.

These constraints show that the Lie algebra of the gauge group equals exactly $\mathfrak{g}_{\max} = \bigoplus_{i=1}^h \mathfrak{gl}(d_k) \oplus \bigoplus_{i=1}^h \mathfrak{gl}(d_v)$, with no cross-head or cross-sector coupling.

Constraint 3: Necessary factorization. After accounting for the permutation σ, the invariance condition requires $V_iW_{O,i} = V_i'W_{O,i}'$ for each head. Since $W_V^{(i)}$ has full column rank, the map $X \mapsto XW_V^{(i)}$ is surjective onto $\mathbb{R}^{n \times d_v}$. For $n \geq d_v$, we can choose V_i with full column rank, forcing $V_i' = V_iC_i$ for a unique $C_i \in \text{GL}(d_v)$. This yields $W_{O,i}' = C_i^{-1}W_{O,i}$.

The same argument applied to the query-key sector establishes that transformations must have the form $(W_Q^{(i)}, W_K^{(i)}) \mapsto (W_Q^{(i)} A_i, W_K^{(i)} (A_i^{-1})^\top)$.

Combining the constraints. Any parameter transformation preserving the MHA function must:

1. Include a head permutation $\sigma \in S_h$ (Constraint 1)
2. Have tangent vectors in $\mathfrak{g}_{\max}$ if continuous (Constraint 2)
3. Factor into independent query-key and value-output transformations (Constraint 3)

These constraints uniquely determine the gauge group to be $G_{\max} = ((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h) \rtimes S_h$, completing the proof. ∎

Appendix B. Proof of Theorem 4

Intuition for $h = 2$. Consider the simplest nontrivial case of two heads. Here the residual symmetry group S_2 has only two irreducible representations: the trivial representation and the sign representation. The nontrivial bispectral energy reduces to measuring how anti-symmetric combinations of head features fail to commute under directional derivatives. This anti-symmetry directly corresponds to the curvature's measurement of how the two heads cannot be independently transformed—they are genuinely coupled in a way that survives all gauge transformations. For larger head counts, this correspondence generalizes through the full representation theory of S_h, with each irreducible component contributing its own measure of inter-head coupling that cannot be gauged away.

B.1. Principal bundle viewpoint and induced actions

Consider the principal $G_{\max}$-bundle $\pi : \Theta \to \mathcal{F}$ with connection 1-form $\Gamma \in \Omega^1(\Theta; \mathfrak{g})$ defined by the Fisher–Rao mechanical connection. Its curvature is $\Omega = d\Gamma + \frac{1}{2}[\Gamma, \Gamma] \in \Omega^2(\Theta; \mathfrak{g})$. For any finite-dimensional (complex) representation $\rho : G_{\max} \to \mathrm{GL}(V_\rho)$ with differential $\rho_* : \mathfrak{g} \to \mathfrak{gl}(V_\rho)$, the associated vector bundle $E_\rho = \Theta \times_{G_{\max}} V_\rho$ inherits a covariant derivative whose curvature is $\rho_*(\Omega)$ (standard functoriality of connections on associated bundles).

After canonicalization, the continuous part is removed and the residual action is the finite permutation group S_h on head indices. We realize the feature map as a section of the associated bundle for the permutation representation restricted to this residual symmetry. Although S_h is discrete, the differential action arises through how the connection couples head-wise responses before quotienting by the continuous gauge; i.e., the variation of f_ℓ along horizontal directions is controlled by the induced connection prior to restricting to the residual discrete action.

B.2. Directional commutator and curvature

Let $u, v \in T_{[\theta_0]}\mathcal{F}$ be horizontal unit vectors and $\tilde{u}, \tilde{v}$ their horizontal lifts at $\theta_0 \in \Theta$. For any associated bundle section s (e.g., a head-feature-derived section),

$$(\nabla_u \nabla_v - \nabla_v \nabla_u) s \;=\; \mathcal{R}(u, v)\, s \quad \text{with} \quad \mathcal{R}(u, v) = \rho_*\big(\Omega(u, v)\big),$$

where ρ_* is the differential of the representation acting on the fiber. This equality follows from torsion-freeness of the Levi–Civita connection on $\mathcal{F}$ and the standard structure equation for associated bundles. Evaluating at the identity fiber representative gives

$$\big(D_u D_v - D_v D_u\big) f_{\ell,\rho}(e) \;=\; \rho_*\big(\Omega_\ell(u,v)\big)\, \mathcal{L}_{\ell,\rho}\, \phi_\ell,$$

where ϕ_ℓ is the canonicalized feature at the base point and $\mathcal{L}_{\ell,\rho}$ is a (bounded) linear map encoding the Jacobian of the feature extraction and canonicalization pipeline with respect to parameters along horizontal directions, projected to the isotypic component ρ.

B.3. Fourier blocks and Schur orthogonality

Write the permutation-feature map $f_\ell : S_h \to \mathbb{C}^m$ and its group Fourier transform $\widehat{f}_{\ell,\rho}$ at irreducible representation $\rho \vdash h$. The selective bispectral energy targeted in nontrivial isotypic components is

$$\mathcal{E}_\ell(u,v) = \sum_{\rho \neq \mathrm{triv}} w_\rho \, \big\|\big(D_u D_v - D_v D_u\big)\widehat{f}_{\ell,\rho}(e)\big\|_{\mathrm{F}}^2,$$

with positive weights w_ρ. By block-diagonalization in the Fourier basis and Schur orthogonality, this quantity is a positive semidefinite quadratic form in the commutator applied to the projected features. Substituting the expression from Section B.2 yields

$$\mathcal{E}_\ell(u,v) \;=\; \sum_{\rho \neq \mathrm{triv}} w_\rho \, \big\|\rho_*\big(\Omega_\ell(u,v)\big)\, \mathcal{L}_{\ell,\rho}\, \phi_\ell\big\|_{\mathrm{F}}^2.$$

B.4. Operator inequality and the constant c_ℓ

Let $\mathcal{A}_\ell$ be the linear operator from the curvature fiber to the concatenated nontrivial isotypic components,

$$\mathcal{A}_\ell :\; X \mapsto \big(\sqrt{w_\rho}\, \rho_*(X)\, \mathcal{L}_{\ell,\rho}\, \phi_\ell\big)_{\rho \neq \mathrm{triv}}.$$

Endow the curvature fiber (a subspace of $\mathfrak{g}$ restricted to layer ℓ) with the Frobenius inner product induced by g_Θ, and the target with the Euclidean/Frobenius product. Then

$$\mathcal{E}_\ell(u,v) \;=\; \big\|\mathcal{A}_\ell\, \Omega_\ell(u,v)\big\|_2^2 \;\le\; \|\mathcal{A}_\ell\|_{\mathrm{op}}^2 \, \|\Omega_\ell(u,v)\|_{\mathrm{F}}^2,$$

and equivalently,

$$\|\Omega_\ell(u,v)\|_{\mathrm{F}}^2 \;\ge\; \lambda_{\min}\big(\mathcal{A}_\ell^* \mathcal{A}_\ell\big)\, \frac{\mathcal{E}_\ell(u,v)}{\|\mathcal{A}_\ell\|_{\mathrm{op}}^2 / \lambda_{\min}\big(\mathcal{A}_\ell^* \mathcal{A}_\ell\big)}.$$

Set

$$c_\ell \;:=\; \lambda_{\min}\big(\mathcal{A}_\ell^* \mathcal{A}_\ell\big) \;>\; 0,$$

which is positive on the nontrivial isotypic subspace under the genericity assumptions (non-degenerate canonicalization Jacobian and FR metric restricted to horizontals). Then

$$\|\Omega_\ell(u,v)\|_{\mathrm{F}}^2 \;\ge\; c_\ell\, \mathcal{E}_\ell(u,v),$$

which is the claimed lower bound in Theorem 4.

Equality conditions. Equality holds when (i) higher-order terms beyond the linearization vanish at $(\theta_0; u, v)$ and (ii) $\Omega_\ell(u, v)$ lies in the minimizing eigenspace of $\mathcal{A}_\ell^* \mathcal{A}_\ell$, equivalently the canonical feature metric aligns with the FR-induced weights so that $\|\mathcal{A}_\ell\|_{\text{op}}^2 = \lambda_{\min}(\mathcal{A}_\ell^* \mathcal{A}_\ell)$.

B.5. Error control for the discrete estimator

Let $K(\varepsilon) = \|\Delta_\square^\ell(u, v; \varepsilon)\|_{\text{F}}/\varepsilon^2$. A standard Baker–Campbell–Hausdorff expansion for the square loop shows $K(\varepsilon) = K(0) + a\varepsilon + O(\varepsilon^2)$ with $K(0) = \|\Omega_\ell(u, v)\|_{\text{F}}$. The two-point Richardson estimator cancels the $O(\varepsilon)$ term; the remaining bias is $O(\varepsilon^2)$. The linearization index

$$\eta(u, v) = \frac{|K(2\varepsilon) - 2K(\varepsilon) + K(\varepsilon/2)|}{K(\varepsilon)}$$

tracks higher-order effects, and filtering by $\eta(u, v) \le \tau$ ensures the commutator approximation dominates.

Appendix C. Additional Proofs and Implementation Details

C.1. Proof of Proposition 3

We prove that the curvature scalars $\kappa_{\ell,i}$ are gauge-invariant by establishing that the curvature form itself transforms equivariantly under the gauge action.

The gauge group $G_{\max}$ acts on (Θ, g_Θ) by isometries, meaning that for any $g \in G_{\max}$ and any tangent vectors $v, w \in T_\theta\Theta$, we have

$$g_\Theta(g_* v, g_* w) = g_\Theta(v, w),$$

where g_* denotes the pushforward of the gauge transformation. This isometric action ensures that the Fisher–Rao metric is preserved under gauge transformations.

The mechanical connection Γ is uniquely determined by the condition that horizontal spaces are Fisher–Rao orthogonal to vertical spaces (the gauge orbits). Since the gauge action preserves both the metric and the vertical distribution, the connection must be $G_{\max}$-equivariant:

$$g^*\Gamma = \text{Ad}_{g^{-1}} \circ \Gamma,$$

where Ad denotes the adjoint action on the Lie algebra $\mathfrak{g}_{\max}$.

The curvature 2-form $\Omega = d\Gamma + \frac{1}{2}[\Gamma, \Gamma]$ inherits this equivariance. For any $\theta \in \Theta$ and $g \in G_{\max}$:

$$\Omega_{g\cdot\theta} = g^*\Omega_\theta = \text{Ad}_{g^{-1}}(\Omega_\theta).$$

The Frobenius norm $\|\cdot\|_F$ on the Lie algebra is defined using the Fisher–Rao metric restricted to vertical directions. Since this metric is invariant under the gauge action, the norm is invariant under the adjoint action:

$$\|\text{Ad}_{g^{-1}}(X)\|_F = \|X\|_F \quad \text{for all } X \in \mathfrak{g}_{\max}.$$

Combining these facts, we obtain

$$\kappa_{\ell,i}(g \cdot \theta) = \|\Omega_{\ell,i}(g \cdot \theta)\|_F = \|\text{Ad}_{g^{-1}}(\Omega_{\ell,i}(\theta))\|_F = \|\Omega_{\ell,i}(\theta)\|_F = \kappa_{\ell,i}(\theta).$$

Therefore, $\kappa_{\ell,i}$ is constant on gauge orbits and descends to a well-defined function on the quotient space $\mathcal{F} = \Theta/G_{\max}$. ■

C.2. Completeness of the full bispectrum

After canonicalization, the residual symmetry is the finite group S_h acting by left translation on head indices. It is therefore natural to view a layer's features as a function $f_\ell : S_h \to \mathbb{C}^m$, obtained by stacking per-head statistics. The left action $(\pi \cdot f_\ell)(\sigma) := f_\ell(\pi^{-1}\sigma)$ models head relabeling, so equivalence reduces to recovery of f_ℓ up to left translation.

Statement. For the permutation group S_h, the *full* bispectrum $B(\sigma_1, \sigma_2)$ evaluated on all pairs $(\sigma_1, \sigma_2) \in S_h \times S_h$ is *generically complete* up to left translation: if $f_\ell, g_\ell : S_h \to \mathbb{C}^m$ have identical full bispectra and, for every irreducible representation $\rho \vdash h$, the channel-stacked Fourier block $\widehat{f_\ell}(\rho)$ has full column rank (equivalently, its Gram matrix is nonsingular on its image), then there exists $\pi \in S_h$ with $g_\ell = \pi \cdot f_\ell$. For $h \neq 6$, all automorphisms of S_h are inner, so "up to automorphism" coincides with "up to translation."

Justification (classical finite-group harmonic analysis). Kakarala [7] established that, for complex-valued functions on a finite group, the triple correlation (bispectrum) determines the function up to translation and group automorphism, under a generic nondegeneracy of Fourier blocks. In our vector-valued setting $f_\ell : S_h \to \mathbb{C}^m$, one stacks channels and applies the same argument componentwise in the Fourier domain: if each nontrivial block $\widehat{f_\ell}(\rho)$ has full column rank, the system of bispectral relations can be solved to recover $\{\widehat{f_\ell}(\rho)\}_\rho$ up to simultaneous conjugation by representation matrices of a single group element, i.e., up to left translation of f_ℓ. Since S_h is finite, no further continuous ambiguity arises. See also standard treatments of finite-group invariants in [15].

Remarks. (i) The rank condition is *generic* and is satisfied whenever head features are not concentrated in a single isotypic component (e.g., independent or sufficiently diverse heads). In practice we verify channel whiteness and nontrivial energy in at least one nontrivial isotypic component (Appendix D.3).
(ii) Completeness here refers to the *full* bispectrum over $S_h \times S_h$. For computational efficiency, one might consider selective subsets, though this can sacrifice completeness.
(iii) For $h = 6$ there is an outer automorphism of S_6; the bispectrum is complete up to that automorphism, which still corresponds to a head relabeling at the level of conjugacy classes and does not affect our canonicalized setting.

C.3. Canonicalization algorithm (pseudo-code)

Algorithm 1 Deterministic canonicalization with stable tie-breaking

1: Balance per-head Q/K Gram matrices (whitening).
2: Orthonormalize V basis per head (QR/SVD).
3: Compute head scores; group heads with gaps below τ_{sort}.
4: Break ties by ℓ_1 norm of vectorized V; then lexicographic order of $\text{vec}(W_O)$; then head index.
5: Return permuted and normalized parameters; record residual permutation action domain as S_h.

C.4. Complexity derivations

Canonicalization costs $O(d_k^3 + d_v^3)$ per head due to SVD/QR. Constructing f_ℓ is linear in $h\, d_v\, n$. The full bispectrum requires $O(h! \cdot h^2)$ operations due to the group Fourier transform over all $|S_h|^2$ pairs, becoming computationally prohibitive for $h > 8$. Holonomy around a square loop requires four directional evaluations (forward/backward around the loop) hence approximately four backprops; reporting over m direction pairs scales linearly in m.

Appendix D. Detailed Empirical Validation

Metric conventions. Unless noted otherwise, correlations are Pearson. The coefficient of variation (CV) is defined as $\mathrm{CV}(X) = \mathrm{std}(X)/\mathrm{mean}(X)$. The reported condition number is for the FR mechanical-connection normal equations $M_\theta \Gamma_\theta(\xi) = b_\theta(\xi)$ with $M_\theta = J_\theta^* G_\theta J_\theta$, namely $\mathrm{cond}(M_\theta)$ in the Euclidean operator norm.

D.1. Curvature Computation via Discrete Holonomy

Directional curvature without explicit metric tensors. Conceptually, curvature on the quotient is defined via the discrete holonomy of a mechanical connection. Let P_{hor} be the horizontal projector and let u, v be unit horizontal directions. For a small step $\varepsilon > 0$, the holonomy around a square loop is $\Delta_\ell^\square(u, v; \varepsilon) = \varepsilon^2 \Omega_\ell(u, v) + O(\varepsilon^3)$, so that $S(\varepsilon) = \|\Delta_\square(u, v; \varepsilon)\|_{\mathrm{F}}/\varepsilon^2$ is a consistent estimator of $\|\Omega(u, v)\|_{\mathrm{F}}$. We use two-step Richardson extrapolation to cancel the $O(\varepsilon)$ term and report the extrapolated value as our curvature estimate. In our implementation, we use the Euclidean gauge-orthogonal connection to instantiate P_{hor} (see Section 3); the discrete holonomy and Richardson extrapolation pipeline remains identical regardless of which connection is used.

D.2. Bound Verification Protocol

Normalization and bound verification protocol. To test Theorem 4, we avoid axis-wise min–max rescaling. For each layer ℓ and direction pair (u, v) we compute the extrapolated curvature magnitude $\widehat{\kappa}_\ell^2(u, v)$ and the directional nontrivial bispectral energy $\widehat{\mathcal{E}}_\ell(u, v)$ in their native units. We then estimate the maximal admissible slope

$$\widehat{c}_\ell \;=\; \min_{(u,v)} \frac{\widehat{\kappa}_\ell^2(u, v)}{\widehat{\mathcal{E}}_\ell(u, v) + \delta}, \qquad \delta = 10^{-10},$$

and verify the bound $\widehat{\kappa}_\ell^2(u, v) \geq \widehat{c}_\ell\, \widehat{\mathcal{E}}_\ell(u, v)$ holds on the vast majority of pairs. For comparability across scales we may multiply both sides by the same positive scalar (e.g., divide by trace of the FR block), which preserves the inequality.

D.3. Bispectral Computation with Equivariance

Equivariant feature map and whitening. Per head i, let $\psi(z_i)$ be the concatenation of head-wise summary statistics (mean, std, range). Define the stacked feature

$$F_\ell = \left[\psi(z_1)^\top \ \cdots \ \psi(z_h)^\top\right]^\top.$$

Under a head permutation $\sigma \in S_h$, F_ℓ transforms by row-permutation. Hence

$$f_\ell(\sigma) = P(\sigma)\, F_\ell$$

with P the permutation representation. Before computing the Fourier blocks and bispectrum we whiten $\psi(z_i)$ across the batch (zero mean, unit covariance per coordinate) to remove amplitude effects that would otherwise confound energy comparisons.

Full bispectrum computation. The full bispectrum requires evaluating $B_\ell(\sigma_1, \sigma_2)$ for all pairs $(\sigma_1, \sigma_2) \in S_h \times S_h$, yielding $(h!)^2$ complex-valued terms. We compute the group Fourier transform $\widehat{f_\ell}(\rho)$ for each irreducible representation $\rho \vdash h$ using the standard character-based projection operators. The computational cost becomes prohibitive for $h > 8$ due to factorial scaling.

Equivariance verification. We verify equivariance through unit tests: for random $\sigma \in S_h$, we confirm

$$\|B_\ell(\text{model}) - B_\ell(\sigma \cdot \text{model})\|_2 < 10^{-12}$$

where $\sigma \cdot$ model denotes the gauge-transformed parameters and B_ℓ is the full bispectrum. This confirms our implementation correctly respects the permutation symmetry.

D.4. Complete Statistical Results

The Richardson ratio reported is $|K(2\varepsilon) - 2K(\varepsilon) + K(\varepsilon/2)|/K(\varepsilon)$; values near 1 indicate stable extrapolation in our step schedule. The condition number is $\mathrm{cond}(M_\theta)$ from the Euclidean mechanical-connection normal equations with $M_\theta = J_\theta^\top J_\theta$, i.e., the gauge-vertical Gram matrix under the Euclidean metric.

Table 3: Complete multi-scale results with Euclidean gauge-orthogonal curvature and whitened bispectral energy.

h	Curvature $\|\Omega\|_F^2$		Bispectral $\mathcal{E}$		Richardson	Condition
	Mean	CV	Mean	CV	Ratio	Number
4	0.058	0.48	17.9	0.36	0.97	3.2×10^3
6	0.066	0.49	26.4	0.33	0.96	4.8×10^3
8	0.072	0.49	35.8	0.28	0.98	6.1×10^3
12	0.081	0.48	52.7	0.25	0.98	9.4×10^3
16	0.089	0.46	64.3	0.25	0.97	1.3×10^4
24	0.097	0.45	76.1	0.24	0.96	2.1×10^4

The Richardson ratios near 0.97 in this table reflect an earlier step-size schedule; the main-text Table 1 reports updated ratios (≈ 1.0 to four decimal places) under our final schedule. Both sets indicate stable extrapolation in the linearized regime. Condition numbers increase with model scale but remain tractable for the iterative solver.

Appendix E. Implementation Methodology and Numerical Validation

This section documents implementation details for computing geometric and algebraic invariants, including numerical safeguards and validation procedures.

E.1. Geometric Computations (Euclidean Gauge-Orthogonal Curvature)

The computation of curvature on the quotient manifold requires careful numerical treatment. As discussed in Section 3, exact gauge invariance renders the Fisher–Rao metric singular along vertical directions, so our implementation uses the Euclidean gauge-orthogonal connection. We solve the linear system $(J_\theta^\top J_\theta)\Gamma = J_\theta^\top \xi$ where J_θ is the vertical map Jacobian. The condition number of this Euclidean system ranges from 3.2×10^3 for 4-head models to

2.1×10^4 for 24-head configurations, enabling stable iterative solution via conjugate gradient with Jacobi preconditioning. Convergence tolerance is set to 10^{-10}, achieved within 50 iterations for all tested configurations.

The discrete holonomy estimator implements Richardson extrapolation with adaptive step sizing. For each directional pair (u, v), we compute $\Delta^{\ell}_{\square}(u, v; \varepsilon)$ at step sizes $\varepsilon_1 = 10^{-4}$ and $\varepsilon_2 = 2 \times 10^{-4}$, chosen to balance truncation and roundoff errors in float64 arithmetic. The stability of the extrapolation is monitored through the Richardson ratio $|K(2\varepsilon) - 2K(\varepsilon) + K(\varepsilon/2)|/K(\varepsilon)$, with values of ≈ 1.0 to four decimal places in our final step-size schedule, indicating stable convergence in the linearized regime (see Appendix D.4 for the full table including earlier-schedule values near 0.97).

E.2. Canonicalization and Residual Symmetry

The canonicalization procedure must be both deterministic and numerically stable to ensure consistent computation of algebraic invariants. Our implementation follows a four-stage process with explicit tolerance parameters:

1. **Query-Key Balancing**: For each head i, compute the Gram matrices $G_Q^{(i)} = (W_Q^{(i)})^T W_Q^{(i)}$ and $G_K^{(i)} = (W_K^{(i)})^T W_K^{(i)}$. Apply simultaneous diagonalization via generalized eigendecomposition with regularization parameter $\epsilon = 10^{-12}$ to prevent numerical instability for near-singular configurations.

2. **Value Orthonormalization**: Apply QR decomposition to each $W_V^{(i)}$ with column pivoting for numerical stability. The resulting orthonormal basis is unique up to column signs, which we fix by requiring the first non-zero element of each column to be positive.

3. **Head Sorting with Stable Tie-Breaking**: Compute sorting metrics s_i for each head based on the Frobenius norm of the value projection. When $|s_i - s_j| < \tau_{\text{sort}} = 10^{-6}$, apply the hierarchical tie-breaking procedure: first by ℓ_1 norm of vectorized V-basis (10^{-9} tolerance), then by lexicographic ordering of $\text{vec}(W_O)$ elements, finally by original head index.

4. **Permutation Tracking**: Record the applied permutation $\sigma \in S_h$ to enable consistent application across all parameter tensors and maintain the relationship between geometric and algebraic computations.

E.3. Validation Protocols

Each implementation component undergoes systematic validation:

Gauge Equivariance Testing: For 100 random gauge transformations $g \in G_{\max}$, verify that computed invariants satisfy $|\kappa(g \cdot \theta) - \kappa(\theta)| < 10^{-10}$ and $\|B(g \cdot \theta) - B(\theta)\|_2 < 10^{-10}$, where B denotes the full bispectrum.

Linearization Index Monitoring: For each direction pair (u, v), compute $\eta(u, v) = |K(2\varepsilon) - 2K(\varepsilon) + K(\varepsilon/2)|/K(\varepsilon)$ and flag cases where $\eta > 0.1$ as potentially violating linearization assumptions.

Numerical Conditioning Assessment: Track condition numbers for the mechanical connection system, canonicalization transformations, and feature whitening matrices. Report warnings when condition numbers exceed 10^6.

Cross-validation with Finite Differences: For a subset of 10 direction pairs per experiment, compare discrete holonomy estimates against high-precision finite difference approximations using step size $h = 10^{-8}$, requiring agreement within relative tolerance 10^{-3}.

E.4. Extended Experimental Protocols

The following details extend the summary in the main text to support full reproduction of our empirical results.

Dataset and Batch Configuration. Synthetic multi-scale and training-dynamics experiments utilize a fixed evaluation batch of 256 sequences of length 128, drawn from the OpenWebText tokenized corpus. The tokenization employs a byte-pair encoding vocabulary of 50,257 tokens, with special tokens for padding, beginning-of-sequence, and end-of-sequence markers. Sequences are selected to maintain diverse linguistic patterns with 40% containing technical content, 30% conversational text, and 30% formal prose. This distribution ensures that attention patterns encounter varied semantic relationships during evaluation. GPT-2 experiments use a separate fixed batch of 16 sequences of length 64 consisting of short English sentences, as specified in Section 6.3. Both batch configurations are held constant across all curvature and bispectrum evaluations within their respective tracks to ensure consistency.

Input embeddings are initialized using Xavier uniform initialization scaled by $\sqrt{d_{\text{model}}}$, with positional encodings following the standard sinusoidal pattern when rotary embeddings are not employed. Layer normalization parameters are initialized with unit gain and zero bias, while attention temperature is fixed at $1/\sqrt{d_k}$ across all experiments.

Model Architecture Specifications. For the multi-scale validation experiments, we systematically vary the number of heads while maintaining proportional scaling of model dimensions. Each configuration maintains fixed query/key dimension $d_k = 64$ and value dimension $d_v = 64$ to isolate the effects of head count on the geometric-algebraic correspondence. The output projection dimension equals $d_{\text{model}} = h \cdot d_v$ to satisfy the architectural constraint for residual connections. Memory requirements scale from 2.1 MB for 4-head models to 75.5 MB for 24-head configurations, enabling single-GPU execution for all experiments.

Training Dynamics Monitoring. The training stability analysis tracks model evolution through 10,000 gradient steps using AdamW with the following hyperparameters. The learning rate schedule implements linear warmup over 500 steps to peak learning rate 5×10^{-4}, followed by cosine annealing to 10^{-5}. Weight decay of 0.1 applies to all parameters except layer normalization and bias terms. Gradient clipping at norm 1.0 prevents instability during early training when attention patterns are uninformative.

Checkpoints are saved at exponentially spaced intervals corresponding to steps 0, 100, 500, 1000, 2500, 5000, and 10000. At each checkpoint, we compute both geometric and algebraic invariants on the fixed evaluation batch, ensuring consistent measurement conditions across training. The same 30 random direction pairs are used at each checkpoint to enable direct comparison of invariant evolution.

Statistical Analysis and Reporting. All reported statistics aggregate over multiple sources of variation. Direction sampling employs stratified random selection to ensure coverage of the parameter manifold. For each layer, we generate 30 direction pairs by sampling

15 pairs from the column space of the Jacobian at randomly selected training points and sampling 15 pairs from random Gaussian directions orthogonalized via Gram-Schmidt.

Confidence intervals are computed using bootstrap resampling with 1000 iterations, reporting the 2.5 and 97.5 percentiles. For bounded quantities like correspondence rates, we apply logit transformation before bootstrapping to avoid boundary effects. The Pearson correlation coefficients reported between geometric and algebraic invariants use Fisher z-transformation for confidence interval construction, with significance testing via permutation tests using 10,000 permutations to avoid parametric assumptions.

E.5. Computational Complexity Analysis

This subsection gives complexity bounds and empirical runtimes for all algorithmic components.

Theoretical Complexity Bounds. The asymptotic complexity of each operation determines the scalability limits of our approach. For canonicalization operations, Gram matrix construction requires $O(h \cdot d_k^2 \cdot d_{\text{model}})$ time with $O(h \cdot d_k^2)$ space, while generalized eigendecomposition scales as $O(h \cdot d_k^3)$ with $O(d_k^2)$ space per head. The QR decomposition for value orthonormalization requires $O(d_{\text{model}} \cdot d_v^2)$ time and $O(d_{\text{model}} \cdot d_v)$ space. Stable sorting with tie-breaking combines $O(h \log h)$ comparison operations with $O(h \cdot d_v^2)$ tie-breaking computations.

For geometric computations, Fisher-Rao metric evaluation scales with the cost of a forward pass through the network, typically $O(n \cdot d_{\text{model}}^2)$ for sequence length n. The mechanical connection solve requires $O(h^2 \cdot (d_k^2 + d_v^2)^{3/2})$ time due to the iterative solver, with space complexity $O(h^2 \cdot (d_k^2 + d_v^2))$ for storing the system matrix. Each discrete holonomy computation requires four backpropagation passes, yielding $O(4 \cdot \text{Backprop})$ time complexity.

The algebraic computations exhibit different scaling characteristics. Feature extraction is linear in the number of parameters, requiring $O(h \cdot n \cdot d_v)$ time with $O(h \cdot m_f)$ space for m_f features per head. Feature whitening involves covariance computation and matrix inversion, scaling as $O(m_f^2 \cdot n + m_f^3)$ with $O(m_f^2)$ space. The full group Fourier transform has factorial complexity $O(h! \cdot h^2)$, which limits practical application to $h \leq 8$.

Empirical Runtime Measurements. On NVIDIA H100 GPUs: full canonicalization ranges from 3.2 ms for 4-head models to 58.2 ms for 24-head configurations; single Fisher-Rao curvature computation scales from 42 ms to 367 ms across the same range; full bispectrum computation grows from 45.6 ms at $h = 4$ to over 8 seconds at $h = 24$, confirming the factorial scaling limitation; mechanical connection solve times range from 8.3 to 112.3 ms.

Memory Requirements and Optimization. Memory is the binding constraint for large models when Fisher-Rao metrics require storing all-parameter gradients. Three strategies reduce peak usage:

Gradient checkpointing reduces memory requirements by recomputing intermediate activations during backpropagation, trading a 40% increase in computation time for 60% reduction in peak memory usage. This trade-off enables analysis of models with up to 48 heads on single H100 GPUs with 80GB memory. Batch-wise accumulation of Fisher-Rao products avoids materializing the full metric tensor, instead computing projections $G_\theta v$ for specific directions v. This reduces memory complexity from $O(p^2)$ where p is parameter count to $O(p)$, enabling scaling to production models.

Mixed precision computation using float32 for invariant calculations and float64 only for critical numerical operations such as canonicalization and mechanical connection solving provides 2× memory savings with negligible impact on correspondence validity. Our experiments show no statistically significant degradation when using mixed precision, with correspondence rates remaining within one percentage point of full float64 results across all tested configurations.

Parallelization Opportunities. Three levels of parallelism reduce wall-clock time. Head-level parallelism enables independent processing of per-head canonicalization and feature extraction across GPU streaming multiprocessors, achieving near-linear speedup up to the number of heads, limited primarily by memory bandwidth for gradient accumulation.

Direction-level parallelism allows concurrent evaluation of curvature for multiple direction pairs, with each pair requiring independent forward-backward passes. This embarrassingly parallel structure scales effectively across multiple GPUs for large-scale invariant analysis. In our experiments, processing 30 direction pairs across 8 GPUs reduces total computation time by a factor of 7.2, with the sub-linear scaling due to communication overhead in gradient synchronization.

Batch-level parallelism in Fisher-Rao computation distributes evaluation examples across devices, with gradient accumulation via distributed reduction. This approach scales to batch sizes of 4,096 samples using 16 GPUs with gradient accumulation over micro-batches of 256 samples per device. The resulting system achieves 12.3× speedup compared to single-GPU execution, limited by the all-reduce communication pattern required for gradient aggregation.

E.6. Computational Requirements and Empirical Runtime

Table 4 presents computation times for the full 7-module pipeline (curvature + canonicalization + bispectral energy + bound verification) on NVIDIA H100 NVL GPUs with PyTorch 2.0 in float64 precision. Times are averaged over 5 seeds with 30 direction pairs each.

Table 4: Computation times for the full 7-module pipeline on NVIDIA H100 NVL. Times are averaged over 5 seeds with 30 direction pairs each for synthetic models; GPT-2 times are per layer (30 pairs).

Configuration	Time per (u, v)	Time (30 pairs)	Peak Memory
Synthetic Multi-Head Attention			
$h = 4$, $d_{\text{model}} = 256$	≈ 0.03 min	≈ 0.9 min	0.9 GB
$h = 6$, $d_{\text{model}} = 384$	≈ 0.08 min	≈ 2.3 min	1.4 GB
$h = 8$, $d_{\text{model}} = 512$	≈ 0.16 min	≈ 4.7 min	2.2 GB
$h = 12$, $d_{\text{model}} = 768$	≈ 0.46 min	≈ 13.7 min	3.1 GB
$h = 16$, $d_{\text{model}} = 1024$	≈ 1.0 min	≈ 29.7 min	5.4 GB
$h = 24$, $d_{\text{model}} = 1536$	≈ 3.6 min	≈ 107 min	5.4 GB
Pretrained GPT-2 Models			
GPT-2 124M (per layer)	—	≈ 14 min	8.2 GB
GPT-2 Medium (per layer)	—	≈ 32 min	12.4 GB

The pipeline scales approximately quadratically with head count. The three-track validation campaign totalled 18.2 hours: Track 1 (multi-scale) 13.2 hours, Track 2 (training dynamics) 2.7 hours, Track 3 (GPT-2) 2.3 hours. Peak memory scales from 0.9 GB to 5.4 GB for synthetic models; GPT-2 models require 8–12 GB. The selective G-bispectrum of Mataigne et al. [10] offers an $O(|G|)$ alternative to the full $O((h!)^2)$ construction.

Appendix F. Reference Implementation and Validation Campaign

This appendix documents the seven-module computational framework and three-track validation campaign used to produce the empirical results in Section 6. All code is implemented in PyTorch 2.0 with float64 precision and validated on NVIDIA H100 NVL GPUs.

F.1. Seven-Module Computational Framework

The implementation decomposes the correspondence validation into seven self-contained modules, each responsible for a specific mathematical operation. This modular design supports correctness through unit testing at each stage.

F.1.1. Module 1: Gauge Algebra Structures

Purpose: Implements the Lie algebra $\mathfrak{g}_{\max}$ of the maximal gauge group $G_{\max} = ((\mathrm{GL}(d_k))^h \times (\mathrm{GL}(d_v))^h) \rtimes S_h$ and the infinitesimal gauge action on MHA parameters. The gauge group characterization follows Wang & Wang [19].

Key Components:

- `MHAParams`: Dataclass storing $(W_Q^{(i)}, W_K^{(i)}, W_V^{(i)}, W_{O,i})_{i=1}^h$ with shape validation.
- `GaugeDirection`: Represents elements $(X_1, \ldots, X_h, Y_1, \ldots, Y_h) \in \bigoplus_{i=1}^h \mathfrak{gl}(d_k) \oplus \bigoplus_{i=1}^h \mathfrak{gl}(d_v)$.
- `vertical_map` (J_θ): Maps gauge directions to parameter tangent vectors via

$$\delta W_Q^{(i)} = W_Q^{(i)} X_i, \quad \delta W_K^{(i)} = -W_K^{(i)} X_i^\top, \quad \delta W_V^{(i)} = W_V^{(i)} Y_i, \quad \delta W_{O,i} = -Y_i W_{O,i}.$$

- `vertical_map_transpose` (J_θ^*): Adjoint operation for mechanical connection computation.

Validation: Unit tests verify that $J_\theta^* J_\theta$ is symmetric positive semi-definite and that the vertical map produces tangent vectors in the kernel of the MHA function's linearization.

F.1.2. Module 2: Fisher–Rao Metric

Purpose: Implements the empirical Fisher–Rao metric g_θ on parameter space via automatic differentiation.

Key Components:

- `fisher_apply`: Computes $G_\theta \xi$ for tangent vector ξ without explicitly forming the full metric tensor, using JVP/VJP operations on a fixed evaluation batch.
- `fr_inner`, `fr_norm`: Inner product and norm under the Fisher–Rao metric.
- `apply_M_operator`: Computes $M_\theta = J_\theta^* G_\theta J_\theta$ for the mechanical connection normal equations.

Degeneracy Discovery: During validation, we confirmed that the Fisher–Rao metric has the vertical subspace V_θ in its null space: for any $\xi \in V_\theta$, we have $g_\theta(\xi, \xi) \approx 0$ (up to numerical precision). This occurs because gauge transformations preserve the MHA function exactly, producing no distinguishable output changes. This degeneracy motivates the Euclidean approach in Module 4.

F.1.3. Module 3: Mechanical Connection (Fisher–Rao)

Purpose: Implements the Fisher–Rao mechanical connection via the normal equations $M_\theta \Gamma_\theta(\xi) = b_\theta(\xi)$ where $M_\theta = J_\theta^* G_\theta J_\theta$ and $b_\theta(\xi) = J_\theta^* G_\theta \xi$.

Note: This module is included for completeness with respect to the theoretical Fisher–Rao formulation in Theorem 4; all empirical results in this paper use the Euclidean connection of Module 4.

Key Components:

- `solve_connection`: Conjugate gradient solver for the normal equations.
- `project_horizontal`, `project_vertical`: Decompose tangent vectors into horizontal and vertical components.

Numerical Issues: As predicted by the Fisher–Rao degeneracy, the matrix M_θ becomes numerically singular with condition numbers exceeding 10^{15}. The CG solver converges in zero iterations because the right-hand side $b_\theta(\xi)$ lies in the null space for any $\xi \in V_\theta$.

F.1.4. Module 4: Euclidean Mechanical Connection and Discrete Holonomy

Purpose: Implements the canonical Euclidean mechanical connection (horizontal spaces as Euclidean orthogonal complements of vertical spaces) and discrete holonomy-based curvature estimation. This approach follows the fiber-bundle treatment of Wang & Wang (2025b).

Key Components:

- `EuclideanMechanicalConnection`: Solves $(J_\theta^\top J_\theta)\Gamma = J_\theta^\top \xi$ via CG with Jacobi preconditioning. Condition numbers range from 3.2×10^3 (4 heads) to 2.1×10^4 (24 heads), enabling stable convergence.
- `generate_horizontal_direction_pair`: Samples random tangent vectors and projects to horizontal space.
- `DiscreteHolonomy`: Computes curvature via the square-loop holonomy estimator with Richardson extrapolation:

$$\|\Omega_\ell(u, v)\|_F \approx \frac{4K(2\varepsilon) - K(\varepsilon)}{3}, \quad K(\varepsilon) = \frac{\|\Delta_\square^\ell(u, v; \varepsilon)\|_F}{\varepsilon^2}.$$

- `perturb_params`: Applies small displacements along horizontal directions for holonomy computation.

Validation: Richardson ratios $|K(2\varepsilon) - 2K(\varepsilon) + K(\varepsilon/2)|/K(\varepsilon)$ are monitored; values near 1.0 indicate stable linearized regime. Our campaign achieved Richardson ratios of 1.0 to four decimal places across all configurations.

F.1.5. MODULE 5: CANONICALIZATION

Purpose: Removes continuous gauge freedom through deterministic normalization, reducing the symmetry group from $G_{\max}$ to the discrete permutation group S_h.

Four-Stage Algorithm:

1. **Query-Key Balancing:** For each head i, compute Gram matrices $G_Q^{(i)} = (W_Q^{(i)})^\top W_Q^{(i)}$ and $G_K^{(i)} = (W_K^{(i)})^\top W_K^{(i)}$. Apply simultaneous diagonalization with regularization $\epsilon = 10^{-12}$.
2. **Value Orthonormalization:** QR decomposition of each $W_V^{(i)}$ with column pivoting; fix signs by requiring positive leading elements.
3. **Head Sorting:** Sort by Frobenius norm of value projection. Tie-breaking hierarchy: ℓ_1 norm of vectorized V-basis (tolerance 10^{-9}), then lexicographic order of $\text{vec}(W_O)$, then original head index.
4. **Permutation Tracking:** Record applied permutation $\sigma \in S_h$ for consistent application across tensors.

Properties: Idempotence (applying twice gives same result) and Lipschitz stability (small parameter changes yield small canonical changes) are verified through unit tests.

F.1.6. MODULE 6: BISPECTRUM COMPUTATION

Purpose: Computes the full S_h bispectrum and directional bispectral energy after canonicalization.

Key Components:

- `extract_head_features`: Computes per-head summary statistics (mean, std, range) from attention activations on evaluation batch.
- `whiten_features`: Zero-mean, unit-variance normalization across batch to remove amplitude effects.
- `compute_triple_correlation`: Full triple correlation $T_\ell(\sigma_1, \sigma_2) = \sum_\tau f_\ell(\tau) f_\ell(\sigma_1\tau) f_\ell(\sigma_2\tau)^*$ over $S_h \times S_h$.
- `compute_bispectrum`: Group Fourier transform of triple correlation into irreducible representation blocks.
- `compute_directional_bispectral_energy`: Second-order finite differences of bispectrum along flattened parameter directions:

$$\mathcal{E}_\ell(u, v) = \sum_{\rho \neq \text{triv}} w_\rho \| D_u D_v \widehat{f}_{\ell,\rho}(e) - D_v D_u \widehat{f}_{\ell,\rho}(e) \|_F^2.$$

Selective G-Bispectrum: For larger groups, one can use the selective G-bispectrum of Mataigne et al. [10], which restricts to an $O(|G|)$ determining set of pairs while preserving discrimination power. Our implementation includes such a determining-set construction, though full bispectrum computations in this paper are restricted to small head counts ($h \leq 8$).

Complexity: Full bispectrum requires $O((h!)^2)$ operations, limiting practical application to $h \leq 8$. For larger models, the validation relies primarily on curvature with bispectral energy computed via the directional approximation.

F.1.7. Module 7: Bound Verification

Purpose: Integrates all modules to verify the theoretical bound $\|\Omega_\ell(u,v)\|_F^2 \geq c_\ell \mathcal{E}_\ell(u,v)$ with proper train/test methodology.

Protocol:

1. Generate n horizontal direction pairs (u_i, v_i) via Module 4.
2. Split into $n/2$ training pairs and $n/2$ test pairs.
3. For each pair, compute curvature $\kappa_i^2 = \|\Omega_\ell(u_i,v_i)\|_F^2$ (Module 4) and bispectral energy $E_i = \mathcal{E}_\ell(u_i,v_i)$ (Module 6).
4. Estimate $\hat{c}_\ell = \min_{\text{train}} \kappa_i^2/(E_i + \delta)$ with $\delta = 10^{-10}$.
5. Evaluate correspondence: fraction of test pairs satisfying $\kappa_i^2 \geq \hat{c}_\ell E_i$.
6. Compute diagnostics: Pearson/Spearman correlations, Richardson ratios, bootstrap confidence intervals.

Output: `BoundVerificationResult` dataclass containing per-pair statistics, aggregate correspondence rate, correlations, $\hat{c}_\ell$ estimate, and Richardson diagnostics.

F.2. Three-Track Validation Campaign

The validation campaign tests the curvature–bispectrum correspondence across three independent experimental settings.

F.2.1. Track 1: Multi-Scale Correspondence

Purpose: Test whether the bound holds consistently across the six model scales $h \in \{4, 6, 8, 12, 16, 24\}$.

Configuration:

- Head counts: $h \in \{4, 6, 8, 12, 16, 24\}$
- Dimensions: $d_k = d_v = 64$, $d_{\text{model}} = 64h$
- Random seeds: 5 per configuration (42, 142, 242, 342, 442)
- Direction pairs: 30 per seed (15 train / 15 test)
- Evaluation batch: 256 sequences × 128 tokens, fixed across all runs

Total Configurations: $6 \times 5 = 30$ independent runs, each with 30 direction pairs.
Runtime: 792.7 minutes (13.2 hours) on NVIDIA H100 NVL.

F.2.2. Track 2: Training Dynamics

Purpose: Verify that the correspondence persists through optimization and is not an initialization artifact.

Model Configuration:

- Architecture: 12 heads, $d_{\text{model}} = 768$, single attention layer with feedforward
- Training data: Synthetic random tokens, vocabulary size 1,000, sequence length 128 (controlled surrogate, not realistic NLP benchmark)
- Optimizer: AdamW, learning rate 5×10^{-4}, weight decay 0.1
- Schedule: Linear warmup (500 steps), cosine annealing to 10^{-5}
- Gradient clipping: Max norm 1.0
- Total steps: 10,000

Checkpoints: Steps 0, 100, 500, 1,000, 2,500, 5,000, 10,000

Per-Checkpoint Validation: 30 direction pairs (15 train / 15 test), full bound verification.

Runtime: 162.1 minutes (2.7 hours) including training and validation.

F.2.3. Track 3: Pretrained GPT-2 Models

Purpose: Test whether the bound holds on pretrained GPT-2 models (124M and 355M parameters), which were not trained for this study.

Models:

- GPT-2 (124M): 12 layers, 12 heads, $d_{\text{model}} = 768$
- GPT-2 Medium (355M): 24 layers, 16 heads, $d_{\text{model}} = 1024$

Layer Selection: Early (layer 0), middle (layer 6 or 12), late (layer 11 or 23).

Per-Layer Validation: 30 direction pairs (15 train / 15 test), full bound verification.

Evaluation Batch: 16 sequences of length 64 consisting of short English sentences, tokenized with the GPT-2 tokenizer. The same fixed batch is reused across all curvature and bispectrum evaluations.

Runtime: 137.7 minutes (2.3 hours) for both models.

F.3. Detailed Campaign Results

F.3.1. Track 1: Per-Seed Multi-Scale Statistics

Table 5 presents per-seed correspondence rates for all configurations, demonstrating the variability across random initializations.

The per-seed variability is highest for $h = 6$ and $h = 12$ (std $\approx 6\%$), where individual seeds occasionally achieve only 83% correspondence. This variability is consistent with the stochastic direction sampling, not with systematic failure of the bound.

Table 5: Track 1 per-seed correspondence rates (%). Seeds indexed 1–5 correspond to random seeds 42, 142, 242, 342, 442.

h	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5	Mean ± Std
4	100.0	96.7	96.7	90.0	96.7	96.0 ± 3.3
6	96.7	100.0	100.0	83.3	96.7	95.3 ± 6.2
8	100.0	96.7	100.0	100.0	100.0	99.3 ± 1.3
12	83.3	90.0	96.7	86.7	100.0	91.3 ± 6.2
16	93.3	96.7	100.0	93.3	100.0	96.7 ± 3.0
24	100.0	100.0	100.0	100.0	96.7	99.3 ± 1.3

Table 6: Track 1 correlation and bound constant statistics with 95% bootstrap confidence intervals.

h	$\lvert\rho\rvert$ (mean ± std)	95% CI	$\hat{c}_\ell$ (mean)	$\hat{c}_\ell$ 95% CI
4	0.189 ± 0.123	[0.08, 0.31]	1.03×10^{-3}	$[2.1 \times 10^{-4}, 2.8 \times 10^{-3}]$
6	0.150 ± 0.067	[0.07, 0.24]	8.40×10^{-4}	$[1.5 \times 10^{-4}, 1.9 \times 10^{-3}]$
8	0.099 ± 0.054	[0.04, 0.17]	1.61×10^{-4}	$[4.2 \times 10^{-5}, 3.1 \times 10^{-4}]$
12	0.171 ± 0.116	[0.05, 0.32]	1.39×10^{-3}	$[2.8 \times 10^{-4}, 2.9 \times 10^{-3}]$
16	0.200 ± 0.142	[0.06, 0.38]	1.81×10^{-4}	$[1.2 \times 10^{-5}, 6.4 \times 10^{-4}]$
24	0.217 ± 0.091	[0.11, 0.33]	1.06×10^{-5}	$[8.7 \times 10^{-12}, 2.9 \times 10^{-5}]$

F.3.2. Track 1: Correlation and Bound Constant Analysis

Table 6 provides the full correlation statistics and $\hat{c}_\ell$ estimates with 95% confidence intervals computed via bootstrap resampling (1,000 iterations).

All correlation 95% CIs exclude values above 0.40, confirming that curvature and bispectral energy carry independent rather than redundant information. The wide $\hat{c}_\ell$ confidence intervals (spanning 1–2 orders of magnitude) reflect the sensitivity of the minimum-ratio estimator to individual direction pairs.

F.3.3. Track 2: Training Dynamics Extended Statistics

Table 7 presents checkpoint-level statistics including curvature and bispectral energy magnitudes. Table 8 provides extended statistics including train/test split correspondence and Spearman correlations.

The train/test split reveals that all correspondence violations occur in the held-out test set, as expected since $\hat{c}_\ell$ is estimated from training pairs. The correlation values remain low throughout training (all $|\rho| < 0.22$), with both Pearson and Spearman correlations showing consistent patterns.

F.3.4. Track 3: GPT-2 Layer-Wise Analysis

Table 9 provides the complete layer-wise statistics for both GPT-2 models.

The extremely small $\hat{c}_\ell$ value for GPT-2 Medium layer 23 (1.56×10^{-15}) indicates that for this configuration, the curvature-to-energy ratio approaches machine precision for some

Table 7: Training dynamics checkpoint results. Correspondence rate computed over 30 direction pairs (15 train / 15 test). Curvature and energy reported in native Euclidean units.

Step	Corr. (%)	$\hat{c}_\ell$	Curvature	Energy	Loss
0	100.0	1.52×10^{-10}	2×10^{-4}	3.56×10^{5}	—
100	100.0	8.21×10^{-10}	1×10^{-4}	2.82×10^{4}	468.0
500	96.7	2.81×10^{-4}	1×10^{-4}	9.43×10^{-2}	21.3
1000	100.0	2.39×10^{-4}	1×10^{-4}	2.98×10^{-2}	8.3
2500	100.0	1.46×10^{-4}	~ 0	3.63×10^{-2}	7.3
5000	96.7	2.87×10^{-8}	~ 0	1.21×10^{2}	7.0
10000	93.3	4.94×10^{-6}	~ 0	1.06	6.9

Table 8: Track 2 extended checkpoint statistics. Train/test correspondence computed on 15-pair splits.

Step	Train Corr. (%)	Test Corr. (%)	$\lvert\rho\rvert$ (Pearson)	$\lvert\rho_s\rvert$ (Spearman)	Loss
0	100.0	100.0	0.021	0.018	—
100	100.0	100.0	0.034	0.029	468.0
500	100.0	93.3	0.187	0.165	21.3
1000	100.0	100.0	0.142	0.121	8.3
2500	100.0	100.0	0.098	0.084	7.3
5000	100.0	93.3	0.211	0.189	7.0
10000	100.0	86.7	0.156	0.138	6.9

Table 9: Track 3 complete GPT-2 layer-wise statistics.

Model	Layer	Train (%)	Test (%)	$\hat{c}_\ell$	$\lvert\rho\rvert$	Time (min)
GPT-2 124M	0 (early)	100.0	86.7	8.28×10^{-4}	0.043	14.7
	6 (middle)	100.0	100.0	9.04×10^{-6}	0.074	13.8
	11 (late)	100.0	93.3	1.40×10^{-6}	0.294	14.1
GPT-2 Med	0 (early)	100.0	86.7	5.74×10^{-4}	0.316	34.6
	12 (middle)	100.0	93.3	4.91×10^{-5}	0.017	30.4
	23 (late)	100.0	93.3	1.56×10^{-15}	0.202	30.0

direction pairs, while still maintaining 93.3% correspondence. This suggests the bound is satisfied but with very different relative scales of the invariants in late layers of larger models.

F.4. Caveats for Result Interpretation

We identify several factors that may affect the interpretation of our empirical results:

Fisher–Rao degeneracy. The theoretical correspondence (Theorem 4) is formulated in terms of Fisher–Rao curvature, but exact gauge invariance renders this metric singular along vertical directions. Our experiments use Euclidean gauge-orthogonal curvature as a computable proxy. While this preserves the inequality form, the absolute magnitudes of curvature differ from the theoretical FR curvature, and the layer-wise constant $\hat{c}_\ell$ is empirically estimated rather than theoretically derived.

Batch dependence. Both curvature and bispectral energy depend on the choice of evaluation batch. Different batches may yield different numerical values, though our experiments use fixed batches within each track for consistency. The correspondence rates we report are specific to these batches; generalization to arbitrary inputs is not guaranteed.

Sensitivity of $\hat{c}_\ell$ estimation. The minimum-ratio estimator $\hat{c}_\ell = \min_{\text{train}} \kappa_i^2/(E_i + \delta)$ is sensitive to outliers: a single direction pair with unusually low curvature or high energy can dramatically reduce the estimate. This explains the wide confidence intervals and multi-order-of-magnitude variation observed across configurations and training stages.

Factorial bispectrum complexity. The full bispectrum requires $O((h!)^2)$ operations, restricting our experiments to small head counts ($h \leq 24$ with approximations). For production models with $h > 24$, we rely primarily on curvature as the validation metric, with bispectral energy computed via directional approximations rather than the full group-theoretic construction.

Linearization assumption. The correspondence is local and linearized: Theorem 4 applies in a neighborhood of a base point where higher-order terms are negligible. Richardson ratios near 1.0 suggest we operate within this regime, but rapid optimization (as in training dynamics) may temporarily violate the linearization assumption, explaining the modest correspondence dips at steps 500, 5000, and 10000.

F.5. Reproducibility and Code Availability

The complete implementation is organized as follows:

- `module1_gauge_algebra.py`: Gauge algebra structures and vertical map
- `module2_fisher_rao_metric.py`: Fisher–Rao metric via automatic differentiation
- `module3_mechanical_connection.py`: FR mechanical connection (for reference)
- `module4_euclidean_curvature.py`: Euclidean connection and discrete holonomy
- `module4_discrete_holonomy.py`: Core holonomy estimation routines
- `module4_diagnostic.py`: Numerical diagnostics and Richardson ratio computation
- `module5_canonicalization.py`: Four-stage canonicalization algorithm

- `module6_bispectrum.py`: Full bispectrum and directional energy
- `module7_bound_verification.py`: Integrated bound verification with train/test split
- `validation_track1_multiscale.py`: Multi-scale correspondence validation
- `validation_track2_training.py`: Training dynamics validation
- `validation_track3_gpt2.py`: Pretrained GPT-2 validation
- `run_validation_campaign.py`: Master orchestrator for complete campaign

All experiments use fixed random seeds for reproducibility. The complete campaign can be reproduced with:

```
python run_validation_campaign.py --track all
```

Quick validation (reduced parameters for testing) is available via:

```
python run_validation_campaign.py --track all --quick
```

Campaign results are saved to JSON files in the `results/` directory with timestamps for provenance tracking.